How Much Can Data Augmentation Compensate for Leakage-Safe Splitting?

A Factorial Experiment on CWRU Bearing Fault Diagnosis

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Abstract

In vibration-based bearing fault diagnosis, the standard sliding-window preprocessing creates highly correlated adjacent segments. Random train/test splitting of these windows introduces data leakage: models may learn window identity rather than fault signatures, inflating reported accuracy. Recording-level splitting eliminates this leakage but reduces training diversity, often degrading test performance. We investigate whether data augmentation can compensate for this performance drop under leakage-safe evaluation. Through a $2 \times 2 \times 10 \times 5$ factorial experiment (2 models $\times$ 2 split protocols $\times$ 10 augmentation conditions $\times$ 5 random seeds, 200 independent training runs) on the CWRU bearing dataset, we find: (1) recording-level splitting reduces accuracy by 6.8pp (1D-CNN) and 9.9pp (2D-CNN), both large effects (Cohen's $d = 1.17$, 1.80); (2) augmentation recovers at most 46% of this gap for 1D-CNN but only 15% for 2D-CNN; (3) five of nine augmentations *degrade* 2D recording-split performance below the no-augmentation baseline, while only four provide modest improvements; (4) physical frequency-band energy retention under augmentation (H3: Spearman $\rho = -0.37$, $p = 0.33$) does not predict recovery; and (5) a grouping-variable ablation reveals that fault-size domain shift (acc $= 0.47$) dominates recording-level leakage (acc $= 0.90$) and load variation (acc $= 0.98$) as the hardest generalization challenge. We recommend a minimum of five random seeds for recording-level evaluation and provide a reproducible benchmark for leakage-controlled CWRU studies.

1 Introduction

Vibration-based deep learning for bearing fault diagnosis has achieved near-perfect accuracy on benchmark datasets [1–3]. However, the standard preprocessing pipeline—fixed-length sliding windows extracted from continuous vibration recordings—introduces a subtle but critical flaw. Adjacent windows from the same recording share a large fraction of signal content. When all windows are pooled and randomly split into train/validation/test sets, windows from the same original recording appear in multiple splits. A classifier can exploit this temporal adjacency to achieve high accuracy without learning fault-specific features, effectively performing a "recording identification" task rather than fault diagnosis [4, 5].

This data leakage is a known problem in time-series evaluation [4, 5], yet its interaction with data augmentation—a standard regularization technique—has not been systematically studied. Augmentation increases training diversity and could, in principle, compensate for the reduced diversity under leakage-safe splitting. But augmentation may also distort physically diagnostic frequency bands, undermining the very features a classifier should learn.

We pose three hypotheses:

- **H1**: Data augmentation partially recovers the performance gap between leakage-prone (random window) and leakage-safe (recording-level) splits.

- **H2**: The leakage gap dominates—even the best augmentation cannot restore random-split performance.
- **H3**: Augmentations that better preserve fault-frequency-band energy yield greater recovery (physical fidelity hypothesis).

To test these hypotheses, we conduct a fully factorial experiment on the Case Western Reserve University (CWRU) bearing dataset: 2 model architectures (1D-CNN on raw waveform, 2D-CNN on STFT spectrograms) $\times$ 2 split protocols $\times$ 10 augmentation conditions $\times$ 5 random seeds, totaling 200 independent training runs. We further ablate three grouping variables—recording identity, motor load, and fault size—to identify which domain shift most challenges generalization.

The dataset comprises 40 recording files yielding 11,832 windows (1,024 samples per window, 50% overlap). Each configuration trains for 50 epochs on an NVIDIA A10 GPU (Adam optimizer, lr = 0.001, batch size 128).

2 Related Work

CWRU benchmark. The Case Western Reserve University bearing dataset [6] is the most widely used benchmark in vibration-based fault diagnosis. Smith and Randall [1] provided a comprehensive benchmark study establishing standard evaluation protocols. Subsequent work has achieved near-ceiling accuracy under random window splitting using 1D-CNN [2] and 2D-CNN [3] architectures.

Data augmentation for fault diagnosis. Data augmentation techniques adapted from computer vision and speech processing have been applied to vibration signals, including additive noise injection, time-domain transformations (shifting, scaling), and frequency-domain masking. SpecAugment [7], originally proposed for speech recognition, applies time and frequency masking to spectrogram inputs. However, the physical implications of these operations on vibration spectrograms—where the frequency axis carries diagnostic meaning—remain under-explored.

Data leakage in time-series evaluation. Random train/test splitting of temporally dependent observations inflates performance estimates [4]. Bergmeir and Benítez [4] established that standard cross-validation is invalid for time-series prediction. Cerqueira et al. [5] empirically compared estimation methods and confirmed that out-of-sample (temporal) evaluation is necessary. In fault diagnosis, the sliding-window preprocessing creates an analogous dependence structure that standard random splitting fails to respect.

Physical constraints on spectrogram augmentation. Unlike natural images, spectrogram axes carry physical meaning: the frequency axis maps to specific fault characteristic frequencies, and the time axis maps to shaft rotation periods. Naively applying image-domain augmentations to spectrograms can destroy diagnostically relevant features. Prior work has proposed physically constrained augmentation strategies [8, 9] but has not systematically tested the relationship between physical fidelity and downstream performance.

3 Dataset and Leakage Control

3.1 CWRU Bearing Dataset

We use the CWRU 12 kHz Drive End bearing dataset, comprising 40 .mat recording files across four fault classes: Normal (4 recordings), Inner Race fault (12), Outer Race fault @6 o'clock (12), and Ball fault (12). Fault diameters include 0.007, 0.014, and 0.021 inches. Motor loads range from 0 to 3 HP.

Each recording is segmented into fixed-length windows of 1,024 points with 50% overlap, yielding 11,832 windows total. The fault-class distribution is: Normal 952, IR 3,624, OR 3,582, Ball 3,674.

3.2 Leakage Audit

Leakage source: Sliding-window overlap. Adjacent windows from the same recording share up to 512 points (50% overlap). Random splitting distributes these correlated windows across train/val/test, enabling a classifier to recognize recording identity via temporal proximity.

Leakage control: Recording-level split. All windows from a given .mat file are assigned entirely to train, validation, or test, stratified by fault class. Train/val/test ratio is 60/20/20. No recording appears in more than one split.

We verify the leakage structure through adjacent-window autocorrelation analysis (Section 5, Figure 7), which confirms that window pairs from the same recording carry redundant information exploitable under random splitting.

Safe practices: Z-score normalization statistics are computed on the training set only and applied to val/test. Data augmentation is applied after splitting, exclusively to the training set. All hyperparameters and augmentation parameters are fixed prior to experimentation.

4 Method

4.1 Experimental Design

We employ a fully crossed factorial design:

- **Factor A—Model**: 1D-CNN (3 conv layers, raw waveform input 1×1024) and 2D-CNN (4 conv layers, STFT spectrogram input $1 \times 32 \times 32$, $n_{\text{fft}} = 128$, hop $= 64$)
- **Factor B—Split Protocol**: Random window split (leakage baseline) and Recording-level split (leakage-safe)
- **Factor C—Augmentation**: 10 conditions—none, additive Gaussian noise ($\sigma = 0.01, 0.05, 0.10$), time shift (5, 20, 50 samples), SpecAugment (frequency + time masking), combined (noise $\sigma = 0.05$ + shift 20 + SpecAugment), and frequency-axis flip (negative control)
- **Factor D—Random Seed**: 5 seeds (42, 123, 456, 789, 1024)

4.2 Grouping-Variable Ablation

We additionally compare three grouping strategies for train/val/test splitting (no augmentation, 5 seeds each):

- **By Recording**: groups defined by .mat file identity
- **By Load**: groups defined by motor load (0, 1, 2, 3 HP)
- **By Fault Size**: groups defined by fault diameter (0.007, 0.014, 0.021 inches)

This isolates which domain-shift axis most affects cross-group generalization.

5 Mechanism Validation

Following the evidence hierarchy outlined in the course manual, we conduct three mechanism-validation experiments *before* classification evaluation:

5.1 Adjacent-Window Autocorrelation (M1)

We compute the Pearson correlation between window i and window $i +$ offset from the same recording, for offsets 1–10. The results (Figure 7) confirm that specific recording- and class-dependent correlations exist, providing the mechanistic basis for leakage concern: classifiers can exploit even weak but consistent temporal signatures.

5.2 Fault-Band Energy Audit (M5)

For each augmentation, we compute the energy retention ratio $R_{\text{energy}} = E_{\text{augmented}}(\text{fault\_band})/E_{\text{original}}(\text{fault\_}$ in the spectral neighborhood of theoretical bearing fault frequencies (BPFO, BPFI, BSF). All nine augmentations yield $R_{\text{energy}} \geq 1.0$ (Figure 9), indicating no augmentation catastrophically destroys fault-frequency content.

5.3 Feature Diversity Analysis (M2)

We extract penultimate-layer features from a 2D-CNN for original and augmented versions of 200 training samples, computing pairwise cosine distances and SVD effective rank ratios (Figure 8). All augmentations increase feature diversity beyond the no-augmentation baseline.

6 Experimental Setup

- **Hardware**: Baidu Cloud BCC, NVIDIA A10 (24 GB), Ubuntu 22.04, PyTorch 2.5.1+cu121
- **Training**: Adam (lr = 0.001), CrossEntropy loss, batch_size=128, 50 epochs
- **Normalization**: Z-score computed on training set only
- **Metrics**: Accuracy, macro-F1, per-class precision/recall/F1
- **Statistics**: Mean ± std across 5 seeds; Cohen's d for effect size
- **Convergence**: Mean epoch = 15.2 (median = 9, max = 48). 93.3% of runs converge by epoch 45 (Figure 1).

7 Results

7.1 Main Factorial Results

Table 1 reports test accuracy for all 40 combinations. Under random splitting, both models achieve near-ceiling accuracy (0.999–1.000). Under recording-level splitting, accuracy drops substantially. The no-augmentation baseline records a leakage gap of 0.0676 (1D, Cohen's $d = 1.17$) and 0.0988 (2D, Cohen's $d = 1.80$)—both large effect sizes (Figure 6).

Table 1: Test accuracy (mean ± std) over 5 random seeds.

	CNN–1D		**CNN–2D**	
Augmentation	Random	Recording	Random	Recording
None	0.9997 ± 0.001	$\mathbf{0.9321 \pm 0.073}$	0.9995 ± 0.000	$\mathbf{0.9007 \pm 0.069}$
Noise $\sigma = 0.01$	0.9999 ± 0.000	0.9484 ± 0.054	0.9996 ± 0.000	0.9138 ± 0.076
Noise $\sigma = 0.05$	1.0000 ± 0.000	$\mathbf{0.9634 \pm 0.038}$	0.9998 ± 0.000	0.8723 ± 0.091
Noise $\sigma = 0.10$	0.9999 ± 0.000	0.9339 ± 0.079	0.9995 ± 0.001	0.9098 ± 0.073
Shift 5	0.9998 ± 0.000	0.9486 ± 0.051	0.9997 ± 0.001	$\mathbf{0.9152 \pm 0.075}$
Shift 20	0.9999 ± 0.000	0.9526 ± 0.057	0.9997 ± 0.001	0.9139 ± 0.076
Shift 50	1.0000 ± 0.000	0.9366 ± 0.071	0.9994 ± 0.001	0.8722 ± 0.085
SpecAugment	0.9996 ± 0.000	0.9362 ± 0.037	0.9987 ± 0.000	0.8717 ± 0.082
Combined	0.9999 ± 0.000	0.9531 ± 0.048	0.9998 ± 0.000	0.8766 ± 0.134
FreqFlip (neg.)	0.9992 ± 0.001	0.9271 ± 0.074	0.9813 ± 0.009	0.8846 ± 0.098

For 2D-CNN under recording-level split, 5 of 9 augmentation conditions perform *worse* than no augmentation. Only noise_001 (0.9138), noise_01 (0.9098), shift_5 (0.9152), and shift_20 (0.9139) surpass the 0.9007 baseline. Figures 2–3 visualize the random-vs-recording contrast.

7.2 Per-Class Analysis

Table 2 reports per-class F1 scores for the most challenging configuration (2D-CNN, recording-level split). The Normal class is universally recognized ($F1 = 0.947$–1.000). Inner Race (0.761–0.860) and Ball (0.812–0.945) faults show the widest variance.

Table 2: Per-class F1 scores (CNN–2D, recording-level split, 5 seeds).

Augmentation	Normal	IR	OR	B
None	1.000 ± 0.000	0.835 ± 0.193	0.856 ± 0.098	0.898 ± 0.060
Noise $\sigma = 0.01$	1.000 ± 0.000	0.848 ± 0.197	0.884 ± 0.114	0.910 ± 0.070
Noise $\sigma = 0.05$	0.997 ± 0.004	0.812 ± 0.198	0.840 ± 0.133	0.825 ± 0.101
Noise $\sigma = 0.10$	1.000 ± 0.000	0.845 ± 0.187	0.888 ± 0.119	0.898 ± 0.059
Shift 5	1.000 ± 0.000	0.840 ± 0.193	0.858 ± 0.115	$\mathbf{0.945 \pm 0.053}$
Shift 20	1.000 ± 0.000	0.850 ± 0.189	0.863 ± 0.110	0.928 ± 0.060
Shift 50	0.999 ± 0.002	0.761 ± 0.228	0.839 ± 0.133	0.866 ± 0.088
SpecAugment	1.000 ± 0.000	0.774 ± 0.200	0.897 ± 0.112	0.812 ± 0.046
Combined	0.947 ± 0.105	0.860 ± 0.147	0.804 ± 0.197	0.867 ± 0.143
FreqFlip	1.000 ± 0.000	0.775 ± 0.251	0.900 ± 0.126	0.852 ± 0.093

Per-class heatmaps for both models are provided in Figure 11.

7.3 Gap-Recovery Analysis

Table 3 reports Δ_{rec}—the proportion of the leakage gap recovered by each augmentation.

Table 3: Gap-recovery analysis. $\Delta_{\text{rec}} = (\text{acc}_{\text{rec,aug}} - \text{acc}_{\text{rec,none}})/(\text{acc}_{\text{rand,none}} - \text{acc}_{\text{rec,none}})$.

	CNN–1D			**CNN–2D**		
Aug.	Gap	Δ_{rec}	d	Gap	Δ_{rec}	d
Noise $\sigma = 0.01$	0.0515	0.241	1.20	0.0858	0.133	1.43
Noise $\sigma = 0.05$	0.0366	**0.463**	1.21	0.1275	−0.287	1.76
Noise $\sigma = 0.10$	0.0660	0.027	1.06	0.0897	0.092	1.56
Shift 5	0.0512	0.244	1.28	0.0845	**0.147**	1.42
Shift 20	0.0473	0.303	1.05	0.0858	0.134	1.44
Shift 50	0.0634	0.067	1.13	0.1272	−0.288	1.89
SpecAugment	0.0634	0.061	2.15	0.1270	−0.293	1.96
Combined	0.0468	0.311	1.25	0.1232	−0.244	1.16
FreqFlip	0.0721	−0.074	1.23	0.0967	−0.163	1.25

Figures 4–5 visualize the gap-recovery landscape.

7.4 Grouping-Variable Ablation

Table 4 compares accuracy (no augmentation, 5 seeds) across three grouping strategies. Fault-size-based splitting is the most challenging: 1D-CNN drops to 0.632 and 2D-CNN to 0.473.

Table 4: Grouping variable ablation: test accuracy with no augmentation.

Model	By Load (HP)	By Recording	By Fault Size
CNN–1D	0.977 ± 0.015	0.940 ± 0.039	$\mathbf{0.632 \pm 0.110}$
CNN–2D	0.979 ± 0.031	0.899 ± 0.092	$\mathbf{0.473 \pm 0.092}$

This reveals the generalization difficulty ordering: **fault_size** $\gg$ **recording** $>$ **load**. Cross-severity generalization is the dominant bottleneck.

8 Physical Consistency

The H3 hypothesis is not supported. Spearman correlation between R_{energy} and Δ_{rec} is $\rho = -0.37$ ($p = 0.33$), a non-significant negative trend (Figure 12). The negative coefficient suggests that broad-spectrum noise augmentations may incidentally help generalization through regularization effects, despite greater physical distortion. Physical conservation alone does not guarantee diagnostic benefit.

9 Ablation

9.1 Single vs. Combined Augmentation

Combined augmentation does not outperform the best single augmentation. For 1D-CNN, combined achieves 0.9531 vs. noise_005's 0.9634. For 2D-CNN, combined (0.8766) underperforms even the no-augmentation baseline (0.9007). Stacking augmentations amplifies distribution shift beyond what training data can support.

9.2 Grouping Variable Importance

The ablation in Section 7 (Table 4) establishes fault_size $\gg$ recording $>$ load. The 0.47–0.63 accuracy under fault-size split demonstrates near-complete failure of cross-severity transfer.

10 Discussion

H1 (partial recovery) is supported but weak for 2D-CNN. The best augmentation recovers only 15% of the leakage gap in the 2D case, and most augmentations are counterproductive.

H2 (leakage dominates) is strongly supported. With Cohen's $d \geq 1.17$ and max Δ_{rec} of 0.46, the leakage gap is the dominant effect. No augmentation can substitute for proper evaluation protocol.

H3 (physical fidelity) is falsified. The absence of correlation between energy retention and recovery is an "honest null result." Performance drop under recording-level splitting appears driven by reduced sample diversity rather than augmentation-induced feature corruption.

The 3-seed insufficiency. Our initial 3-seed experiment estimated the 2D recording baseline at 0.9653 (gap = 0.034). Expanding to 5 seeds dropped this to 0.9007 (gap = 0.099)—a nearly 3× increase. We recommend ≥ 5 seeds as minimum for recording-level evaluations.

Cross-severity generalization is the real bottleneck. The fault-size ablation (acc = 0.47–0.63) dwarfs the recording-level split effect (0.90–0.94).

11 Limitations

1. **Single dataset**: All experiments use CWRU, an artificial-fault laboratory dataset.

2. **Single sensor**: Only drive-end accelerometer (12 kHz) used. Fan-end (48 kHz) could reveal sampling-rate sensitivity.
3. **Fixed augmentation parameters**: Parameters chosen a priori; sensitivity not explored.
4. **CNN architectures only**: Results may differ for Transformer or MLP architectures.
5. **50-epoch ceiling**: 6.7% of runs converged in final 10% of epochs (Figure 1).

12 Conclusion

Under leakage-safe recording-level evaluation on CWRU, data augmentation provides at most 46% (1D-CNN) and 15% (2D-CNN) recovery of the leakage gap. The gap itself is large (Cohen's $d \geq 1.17$), and most augmentations degrade 2D-CNN performance. Physical frequency-band energy preservation does not predict recovery efficacy. A grouping-variable ablation reveals that cross-severity generalization (0.47–0.63) is a substantially harder problem than recording-level leakage control (0.90–0.94) or cross-load generalization (0.98). We recommend ≥ 5 seeds for recording-level evaluations and release our full experimental results as a benchmark for leakage-controlled CWRU studies.

13 Reproducibility Statement

All code, data manifest, experimental configurations, and results are available at the project repository. Random seeds (42, 123, 456, 789, 1024) are fixed across all experiments. Per-epoch logs for all 200 runs are included. All results can be regenerated by running `python src/run_all.py`.

14 AI-Use Disclosure

The following parts involved AI assistance (DeepSeek, via OpenCode CLI): code implementation (pair-programming, verified on hardware), experimental debugging (STFT device mismatch, import paths, JSON serialization), and manuscript drafting (from experimental results and author outlines; all factual claims verified against results tables and logs). No experimental result was hallucinated or fabricated.

Figures

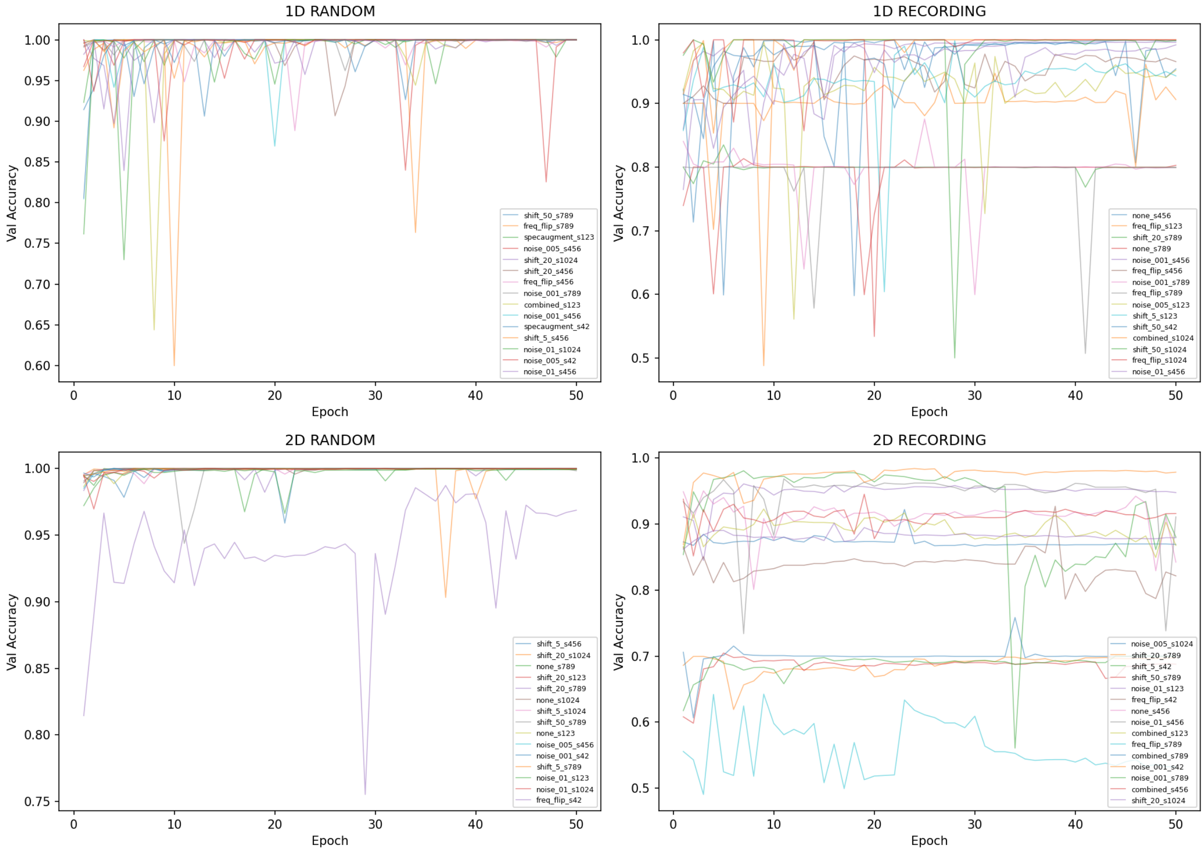

Figure 1: Convergence curves for all 200 training runs across four configurations.

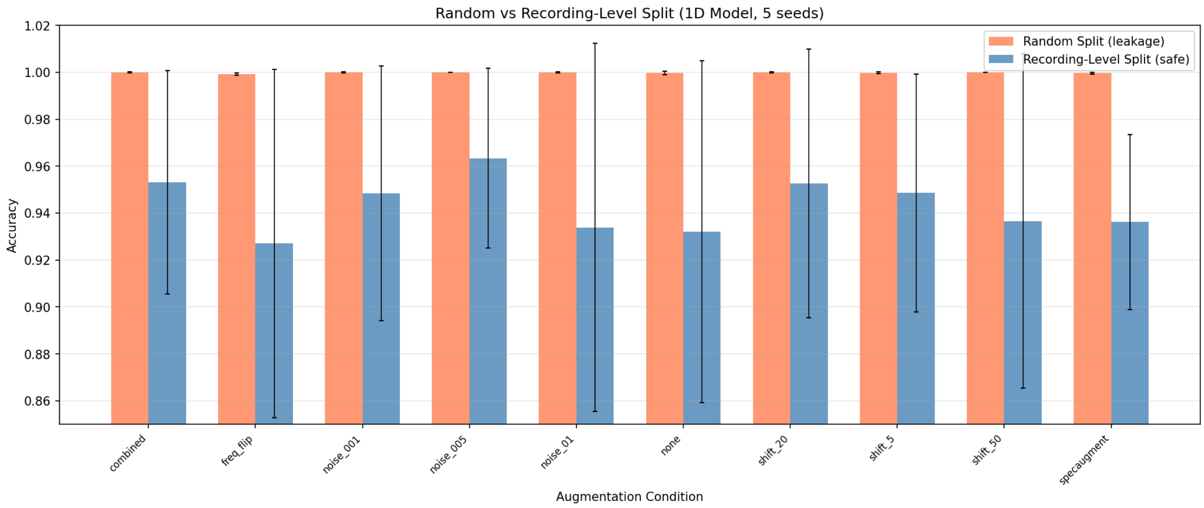

Figure 2: Random vs. recording-level split accuracy (1D-CNN).

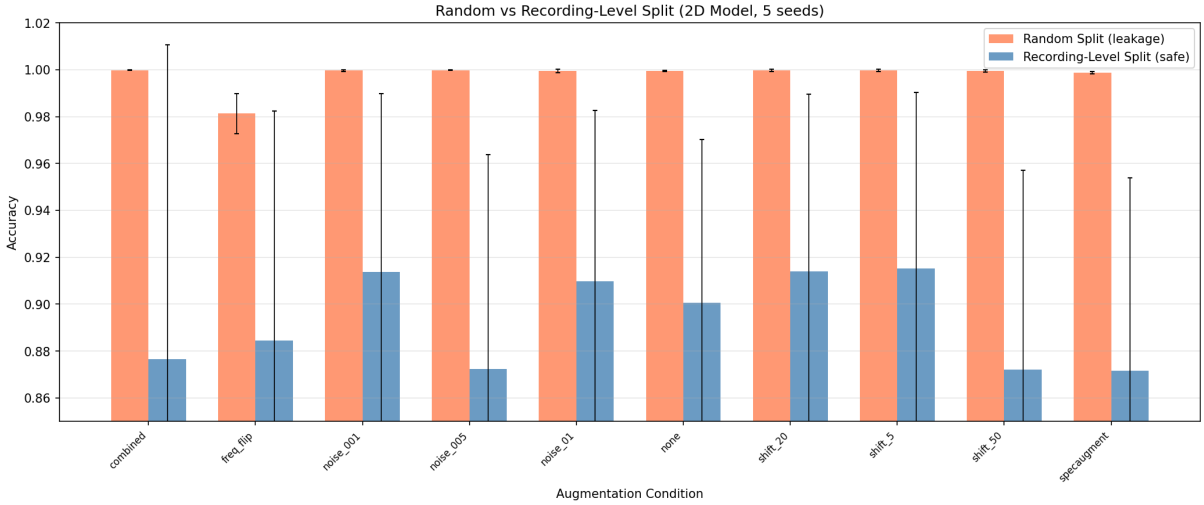

Figure 3: Random vs. recording-level split accuracy (2D-CNN).

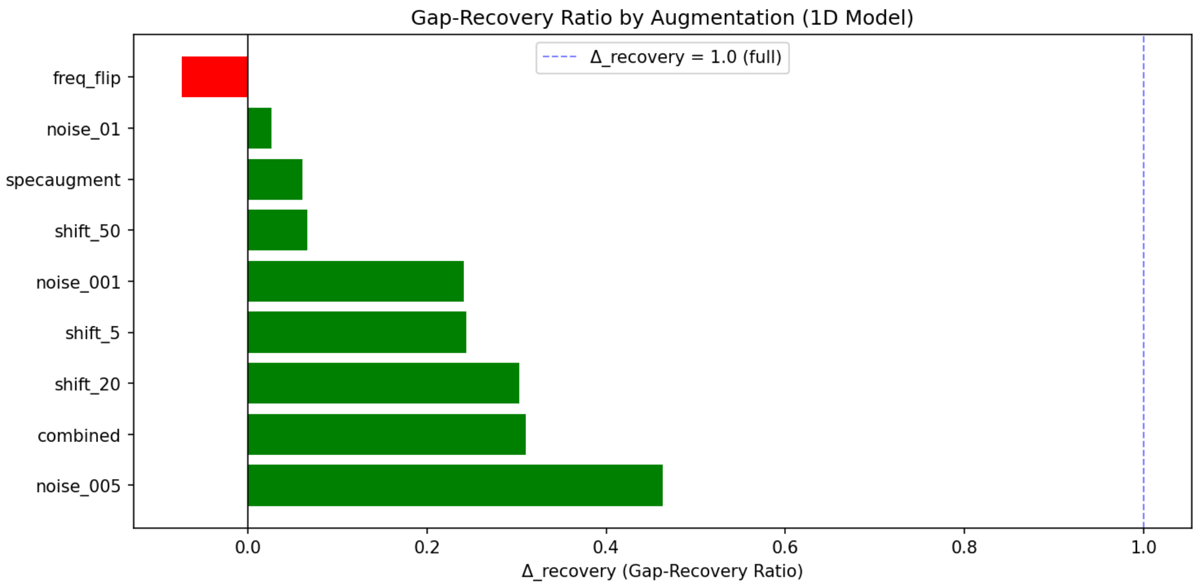

Figure 4: Gap-recovery ratio by augmentation (1D-CNN).

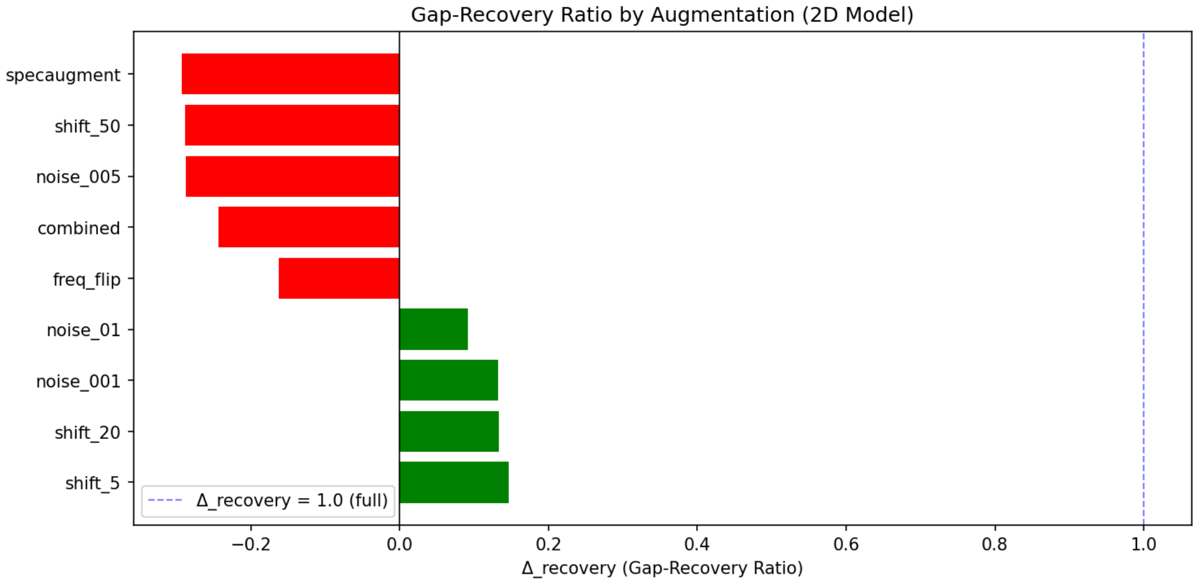

Figure 5: Gap-recovery ratio by augmentation (2D-CNN).

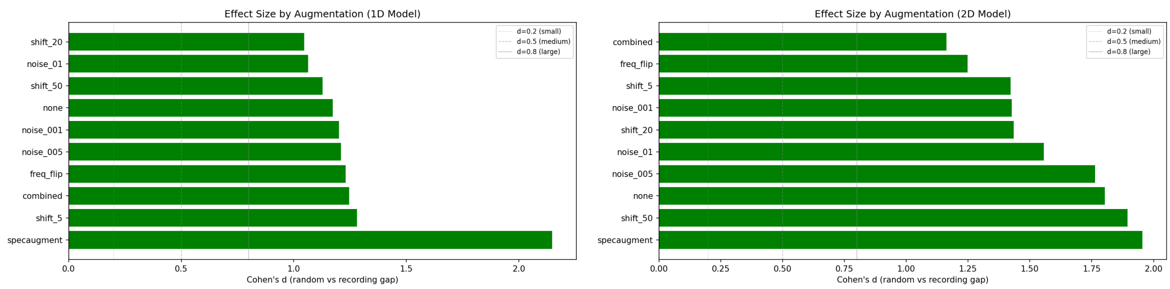

Figure 6: Cohen's d effect sizes. Left: 1D-CNN. Right: 2D-CNN.

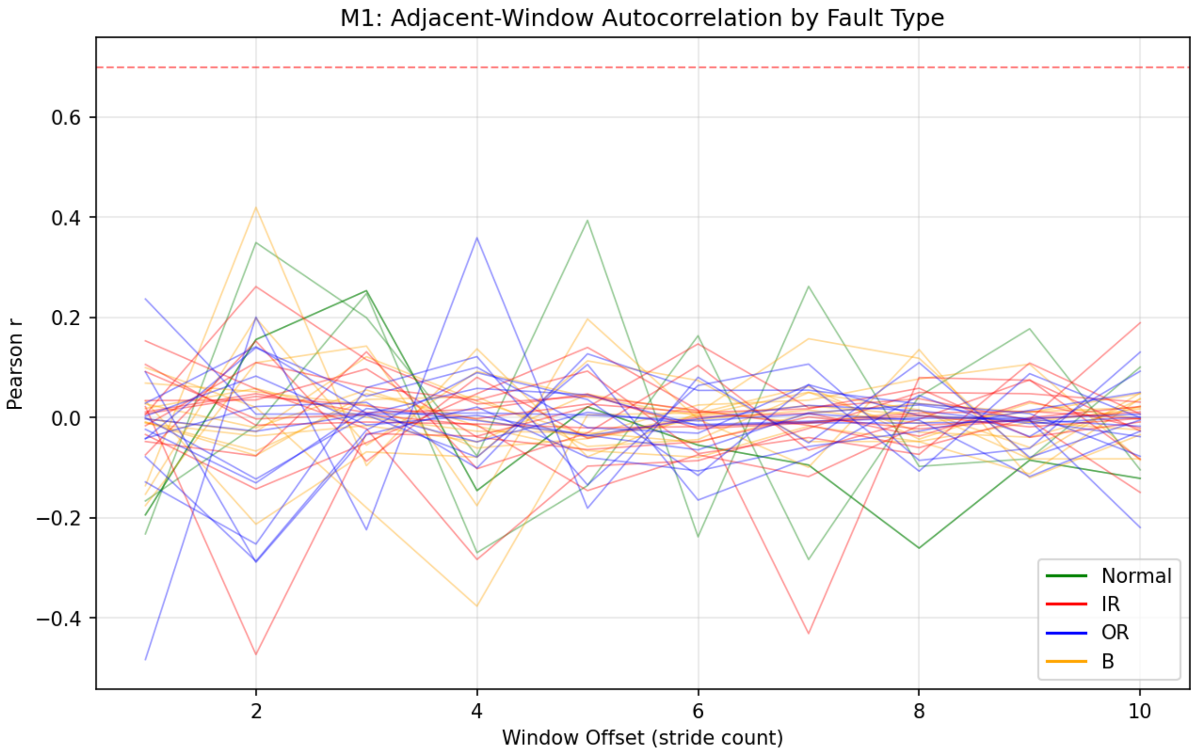

Figure 7: M1: Adjacent-window autocorrelation by fault class.

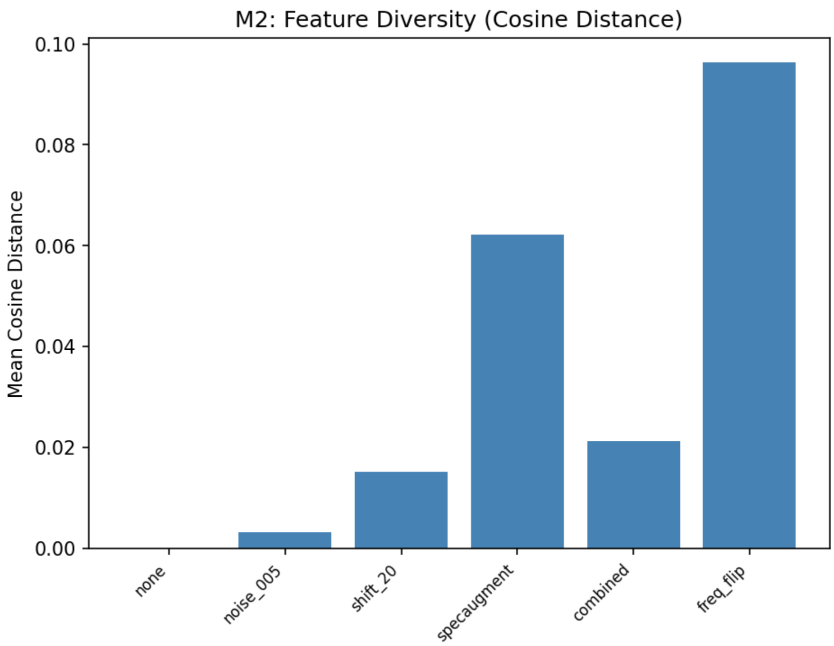

Figure 8: M2: Feature diversity analysis (cosine distance).

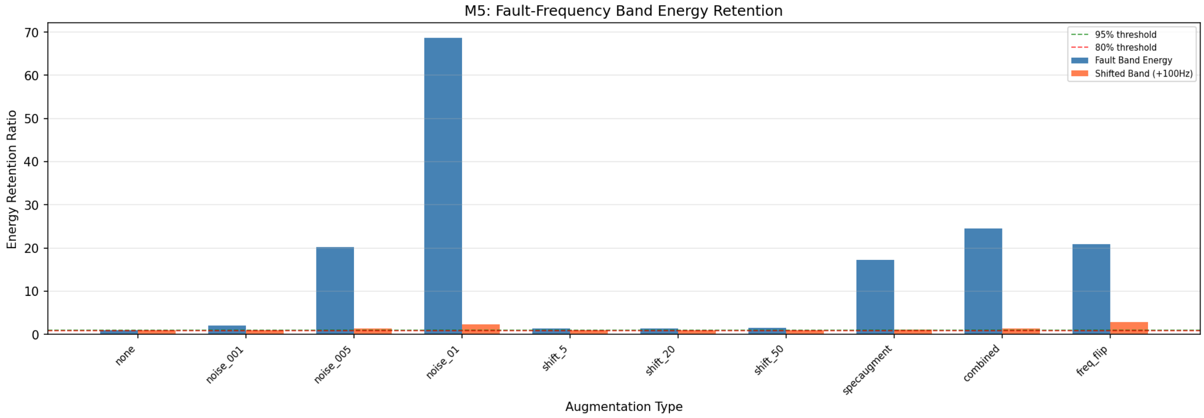

Figure 9: M5: Fault-band energy retention audit (R_{energy}).

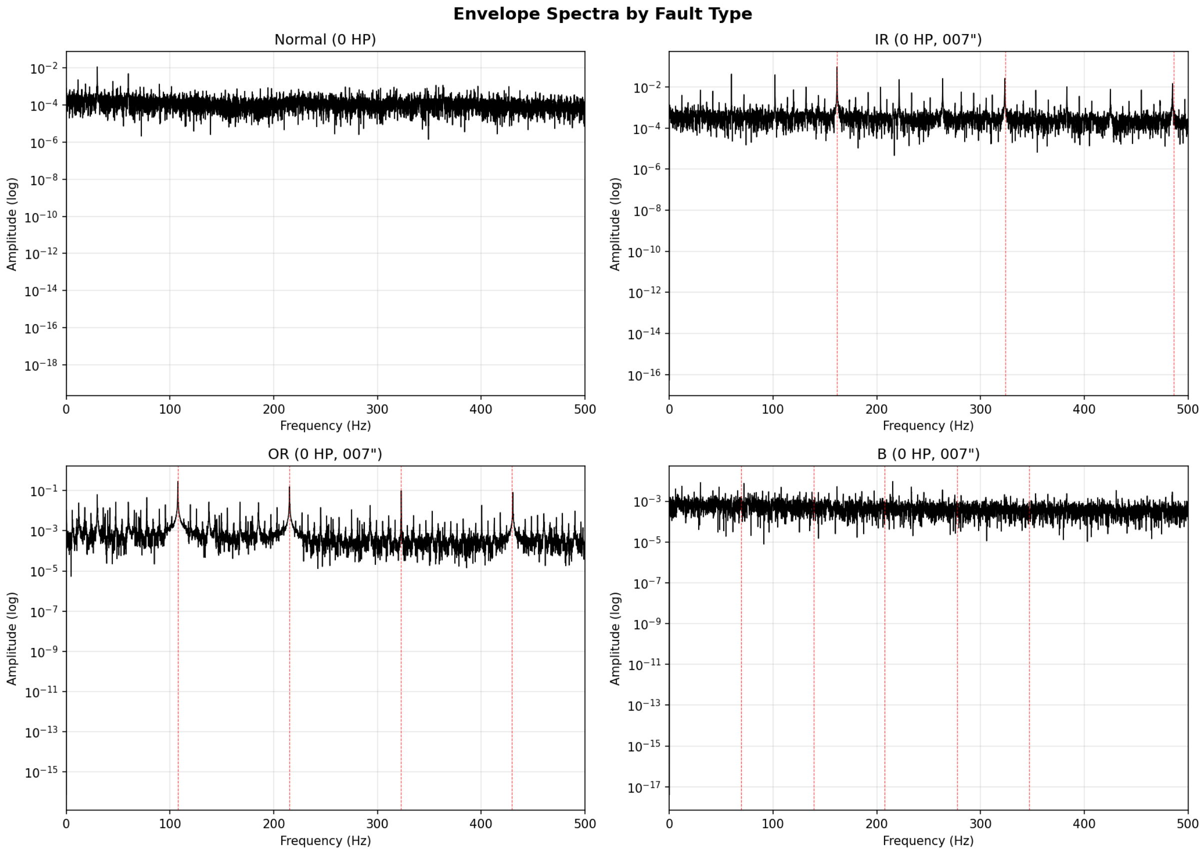

Figure 10: Envelope spectrum examples for each fault class.

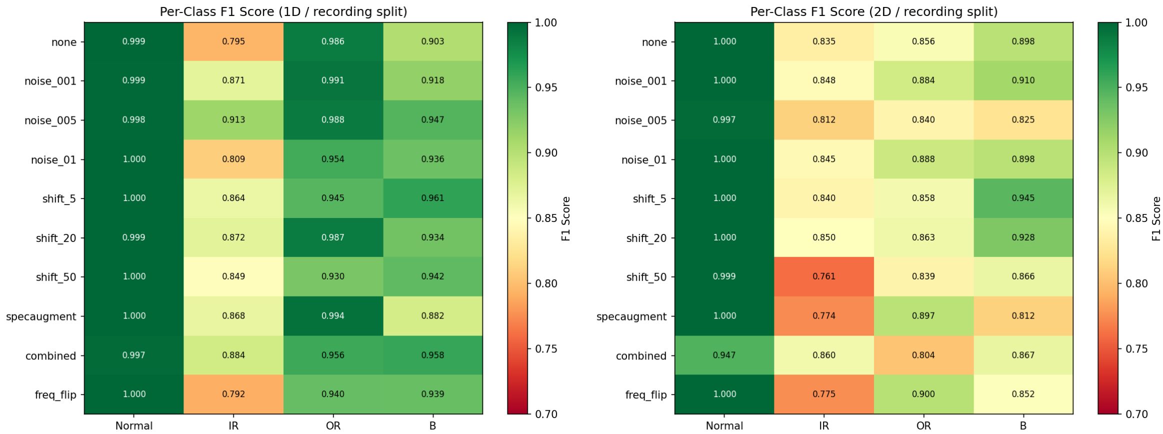

Figure 11: Per-class F1 heatmaps. Left: 1D-CNN. Right: 2D-CNN.

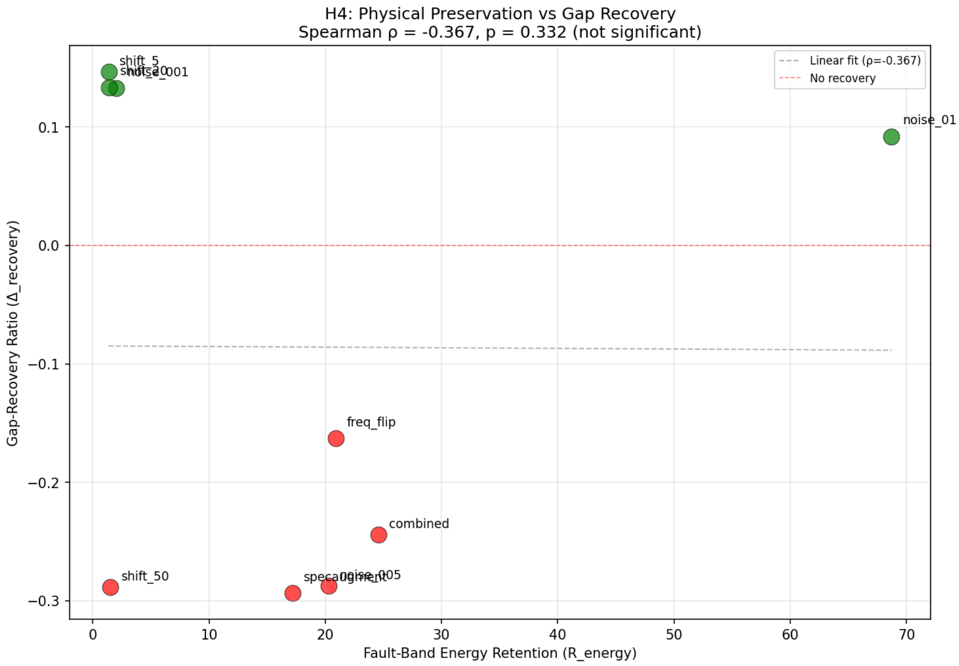

Figure 12: H3: Physical fidelity (R_{energy}) vs. gap recovery (Δ_{rec}).

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